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Addressing the gaps between higher education research and management education

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A R T I C L E I N F O

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A B S T R A C T

The purpose of this paper is to address the gaps between the higher education research, and management education. These two fields seem to have developed with surprisingly little interaction, an impression that is reinforced by citation count. As a guide to redressing this situation, this paper explores an important subset of the higher education literature, sometimes called the "student learning" research, and its implications for management education. This research underpins government-sponsored surveys of program quality in Britain and Australia. I argue that it offers new insights into course structure, pedagogy and assessment. If we want students to understand, evaluate and critically analyze management situations, it is counterproductive to use pedagogies or assessments that higher education research has shown to actually undermine these objectives. The student learning literature offers guidelines for course design and simple approaches to course evaluation. Interventions based on this research have shown remarkable improvements in learning outcomes. Implications for practice and opportunities for research are noted throughout the paper.

1. Purpose

Management education is largely taught in university business schools and it would be reasonable to assume that research in management education would be a subset of, or at least closely related to, higher education research. However, a study of the literature generates the impression that advances in one are not always taken up by the other. For example, the use of variation theory has shown dramatic improvements in learning outcomes (Åkerlind, 2015) and although these have been reported in the higher education literature, there are no studies of its use in this journal or in other journals in management education. Worse, there are no reports of the use of variation theory in management education practice, in spite of the "dramatic" improvements in learning outcomes, reported by Åkerlind. The potential value of variation theory to enhance management education, especially in the design of case method courses, is yet to be realized.

Variation theory is just one of several examples where developments in higher education research have had little or no impact on management education research or practice. The findings and the methods of higher education research have a great deal to offer management education. For example, the guidelines for course design and the simple, research-based techniques for evaluating course quality, developed by higher education researchers, provide proven approaches for course improvement. as well as offering new areas for research.

The rather limited interaction between higher education research and management education research, is also reflected in simple

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citation counts. For example, Ference Marton, identified as one of the most influential authors in higher education (Kandlbinder, 2013), has been cited over 750 times in the top four journals in higher education, an average of almost 200 per journal. By contrast, he has only been cited 43 times in the top four journals in management education, an average of under 11 citations per journal. This includes just 17 citations in this journal.

Taken together, the lack of any studies using variation theory in management education journals, the limited use of validated surveys available to research pedagogies and course design in management education, and the limited interaction with higher education research, by citation count, suggests that there may be value in extending the use of higher education research in management education. Hence the purpose of this paper is to explore the higher education research and to identify the most promising opportunities to apply the methods and the findings from that research to management education.

2. Introduction

Identifying the aspects of the higher education literature that are of most use to management educators, has resulted in a focus on a subset of this literature which Biggs calls the "student learning research" (Biggs, 1999, p. 59; Biggs et al., 2022, p. 22). This research is based squarely on the student experience of learning. Starting in the 1970s, the student learning research has developed over subsequent decades to dominate the higher education research literature (Kandlbinder, 2013). The early findings from the student learning research were used as the basis for government mandated surveys of program quality in both Britain (the National Student Survey) and Australia (the Course Experience Questionnaire and the Student Experience Survey) (Buckley, 2012; Ramsden, 1991). However, more recent developments have taken its contributions far beyond its use in institution level surveys of program quality. These developments include studies of the factors that influence learning, and these studies have led, in turn, to guidelines for course design and delivery, and to simple surveys for evaluating changes in courses and for comparing alternative pedagogies. More recently, the student learning research has been extended to develop variation theory (Åkerlind, 2015).

Some of the improvements in student learning outcomes, reported in the student learning literature, were quite remarkable. For example, in one study, it was found that only 26% of students achieved a good understanding of basic accounting concepts in a traditional course, whereas 85% of students achieved a good understanding when relatively minor changes were made to the same course (Rovio-Johansson & Lumsden, 2012). The findings from this study and from other studies in the student learning literature are a rich resource for management educators.

Similarly, the research methods and surveys, developed as part of the student learning research, can be used and adapted by management education researchers. Phenomenography, a qualitative research method, was actually developed as part of the early studies that laid the groundwork for the student learning research (Marton, 1981, 1988). Short surveys based on the student learning research, are available to compare pedagogies (see Biggs et al., 2001; Entwistle et al., 1997, 2013) or to investigate approaches to teaching (Trigwell & Prosser, 2004; Trigwell et al., 2005).

The student learning research has received little attention in the management education literature. The most cited authors in higher education research, identified by Kandlbinder (2013) – Marton, Biggs, Entwistle, Ramsden, Trigwell and Prosser - are seldom mentioned in the management education literature. It is interesting that there have been calls for consideration of the broader education literature in management education, especially calls for more student-centered approaches, but few (if any) of them has mentioned the student learning literature or cited any of the leading writers in this field (see, for example, Brown et al., 2013; Frost & Fukami, 1997; Rynes & Brown, 2011; Whetten, 2007). One notable exception is a paper by Hrivnak (2019), in which he argued for management educators to pay more attention to course design, using Biggs' concept of constructive alignment as a framework.

Constructive alignment (Biggs, 1996; Biggs et al., 2022) could be considered an overarching framework within which the student learning research can be used to guide choices of topic sequencing, pedagogy, learning activities and assessment. Constructive alignment is a deceptively simple concept. In a nutshell, Biggs argues that all aspects of a course should be aligned to produce the desired learning outcomes (Biggs, 1996; Biggs et al., 2022). If we want to teach students how to critically analyze a management situation, it is of little use to use a course structure, pedagogy or assessments that promote rote learning of course content, with little or no appreciation of critical analysis. The problem with such a simple concept is the implementation. How do we design a course that is "constructively aligned"? Biggs et al. (2022) suggest using findings from the student learning research.

After a brief introduction to the student learning research, I discuss its potential contributions to management education. Having used this research to guide my own practice as a management educator over the last thirty years, I use my own experience to illustrate its application in management education.

3. The student learning research

This research was started in Sweden in the mid 1970s, and was further developed in Britain, Hong Kong and Australia. The work has continued, especially in Europe, Scandinavia and Australia, with some of the original researchers still publishing in this field as recently as 2022.

In their seminal works, Marton and Säljö (1976a, 1976b) found that when faced with a learning task, some students tried to memorize the material for reproduction in assessment. Others took a quite different approach, engaging more meaningfully with the material, seeking to understand, evaluate and critique. These students made connections with prior knowledge or experience to appreciate the significance of the material, and to situate it in their own mental models of the field (see the quotes from students in Marton & Säljö, 2005, pp. 44–45). The original researchers used the terms "surface-level" and "deep-level" to describe these approaches to learning, and the terms were later abbreviated to "surface" and "deep". These labels, together with the short descriptions "learning with intent and strategies to memorize" and "learning with intent and strategies to understand" respectively, have led to the concept of approaches to learning being misunderstood by some academics as a simplistic dichotomy – memorizing versus understanding. It should be appreciated that the concept of a deep approach to learning is much more than just trying to understand. It includes all the higher stages of Bloom's taxonomy – understanding, applying, analyzing, evaluating, and even creating (Anderson & Krathwohl, 2001; Bloom et al., 1956; Krathwohl, 2002). A summary of the characteristics of the two basic approaches to learning is shown in Table 1. This makes clear the significance of approaches to learning as identified in the early years of the student learning research. A later study by Leung and Kember (2003) found a clear relationship between a deep approach to learning and "critical reflection", or critical thinking. In summary, if we want students to move beyond rote learning through the higher stages of Bloom's taxonomy, and to develop the ability to think critically, then we have to design and deliver courses so as to foster a deep approach to learning. As explained in the remainder of this paper, the student learning research offers guidance on how we can do this better. To complement this guidance, researchers have developed simple surveys to assess the extent to which course design, delivery and assessment affect student approaches to learning. The extent to which a course promotes a deep approach and discourages a surface approach, is taken as a measure of course quality.

Entwistle and Ramsden (1983) identified a third approach to learning. Rather than being different to the original two approaches, this third approach, called an achieving or strategic approach, involves switching between a surface approach and a deep approach, as the student believes necessary in order to maximize grades. This is important because it indicates that students' approaches to learning often can and do change. Within the constraints of their own capabilities, students adopt different approaches in response to the context (Entwistle & Ramsden, 1983; Ramsden, 2005). Thus, approaches to learning can be influenced, perhaps even shaped, by the practices of the institution and the individual teacher.

The student learning research has identified factors that influence student learning. Based on this research and especially on the work of Biggs et al. (2022 and Trigwell and Prosser (2020), these factors can be summarized under the following headings.

1. Students' conceptions of learning
2. Teachers' conceptions of teaching
3. Students' motivation to learn
4. Course design and delivery, including
   a. Pedagogies
   b. Sequencing of learning activities
   c. Workload
   d. Assessment

When considered under these categories, it becomes apparent that most of these factors are under the control of, or at least strongly influenced by, the teacher and the institution. Very few are completely outside their control. Management educators must accept that they and their institutions have the primary responsibility to promote a deep approach to learning on the part of their students. In other words, it is our job to promote understanding, application, evaluation, synthesis, reflection, and critical thinking, by establishing the conditions necessary for students to adopt a deep approach to learning. The student learning literature provides the basis for both practice and research in this endeavour.

4. Students' conceptions of learning

Marton et al. (1993) built on previous work to identify student conceptions of learning (Marton & Säljö, 2005). These were.

1. Increasing one's knowledge,
2. Memorizing and reproducing,
3. Memorizing, for subsequent application, of facts, methods, etc.,
4. Understanding,
5. Seeing things in a different way, and

Table 1
Characteristics of approaches to learning (adapted from Southern Cross University Center for Teaching and Learning, n.d. and Entwistle & Peterson, 2004).

Deep approach to learning	Surface approach to learning
Learning with intent and strategies to achieve a thorough understanding of the subject	Learning with intent and strategies to memorize for reproduction in assessment
Actively seek to understand the material/the subject	Try to learn in order to repeat what they have learned
Interact vigorously with the content	Memorize information needed for assessments
Make use of evidence, inquiry and evaluation	Take a narrow view and concentrate on detail
Relate new ideas to previous knowledge	Fail to distinguish principles from examples
Tend to read and study beyond the course requirements	Tend to stick closely to the course requirements
Are motivated by interest	Are motivated by fear of failure, or by the desire to achieve a passing grade with minimum effort.

6. Developing as a person.

Van Rossum and Schenk (1984) found that students holding any of the first three conceptions of learning in the list above, tended to adopt a surface approach to learning, while students who held any of the last three conceptions, tended to adopt a deep approach. These conceptions are generally shaped by prior experience in learning situations. Students who have completed previous courses by memorizing material from lectures or textbooks and then sitting for examinations, which are basically tests of short-term memory, often think of learning as memorizing. Those who have completed a course that required a more critical approach, tend to hold one of the last three conceptions of learning in the list above.

Of course, there is always the opportunity to challenge student conceptions of learning. While a lecture on the topic might seem to be the quickest and easiest way to do this, the evidence is that lectures generally do not change conceptions (Trigwell et al., 1999). I have found that a more interactive approach is usually more effective. Simply asking students to write down what they understand by the word "learning", getting them to compare their ideas in small groups and then writing their responses on a whiteboard, can be quite enlightening. Generally, the student responses can be grouped into the first five of the conceptions listed above. A short class discussion can then lead to the inclusion of the last conception. This exercise challenges many students and opens the way to a deep approach to learning. However, it must be emphasized that this approach has not been rigorously researched.

Regardless of how well such an exercise is done, it must always be complemented by the remainder of the course. In Biggs' terms, everything in the course must be aligned to produce the desired outcome. My own experience indicates that when students are faced with learning exercises and assessments that require understanding and application, the majority usually adopt a deep approach to learning. This perception is confirmed in studies by Pang and Marton (2003, 2005), Rosier (2022) and Rovio-Johansson and Lumsden (2012), in which between 40% and 60% of students apparently switched from a surface approach to a deep approach, in response to changes in course design. There is scope for extensive research, building on these findings in management education.

5. Teachers' conceptions of teaching

Trigwell et al. (1999) identified a range of conceptions of teaching held by university teachers. Their studies found that teachers who had more of an "Information Transmission – Teacher Focused" (ITTF) approach to teaching, at one end of the range, tended to produce a surface approach to learning in their students, whereas teachers who had more of a "Concept Changing – Student Focused" (CCSF) approach tended to produce a deep approach in their students. It should be noted that the subjects in this study were teachers and students in first year university physics and chemistry courses. While there have been some studies of the beliefs and intentions of business academics regarding teaching (Mesny et al., 2021; Rienties et al., 2015), there have been no studies linking teacher conceptions with measures of learning outcomes in business courses. If the relationship found by Trigwell et al. holds for business teachers, this could be the basis for rethinking recruitment, development and promotion of business academics. One concerning finding from the work of Reinties et al. (2015) is that business academics seemed to be more resistant to changing their beliefs about teaching than academics from other disciplines. The authors of this study suggested that this resistance could be due to a combination of large classes and the pressure to research and publish in their discipline areas.

Understanding these factors, and their effects on learning outcomes, is important for the management of business schools, but research in these areas is sadly lacking. However, the student learning research provides both the surveys and the criteria for such research. Building on earlier phenomenographic studies, Trigwell and Prosser published details of their "Approaches to Teaching Inventory" (ATI), and a revised version of the ATI, used to measure the conceptions of teaching held by university teachers (Trigwell & Prosser, 2004; Trigwell et al., 2005). These inventories can be used in conjunction with measures of approaches to learning, such as the Revised Study Process Questionnaire (Biggs et al., 2001), as the basis for research studies or for organizational development programs in business schools.

If we are to align all aspects of a course to achieve the desired learning outcomes, then the teachers' conceptions of teaching are the starting point for this process. Hence a research-based appreciation of conceptions of teaching and their effects on learning, is crucial for effective management education.

6. Students' motivation to learn

A study by Fransson (1977) found that student interest in the subject matter (intrinsic motivation) was an important, but not sufficient condition for students to adopt a deep approach to learning. Uninterested students were unlikely to adopt a deep approach. However, even the deep approach favored by interested students could be undermined by the anxiety produced by various forms of extrinsic motivation (time pressure, a test, or the need to do a formal presentation on the subject). Indeed, in Fransson's experiment, the only condition that produced a deep approach in all his subjects was intrinsic interest in the subject and a lack of anxiety produced by extrinsic pressures.

Fransson's findings are supported by Biggs (1987), Biggs et al. (2022, p. 24), Boud (1995) and Ramsden (2003, p. 80). Boud in particular, argues that "we must confront the ways in which assessment tends to undermine learning" (1995 p. 35). It would seem from this that management educators should pay particular attention to stimulating students' interest in the subject and to minimizing the effects of other factors (especially anxiety) that undermine learning. Pedagogies such as the case method, simulations, role plays and industry-based projects are believed to stimulate interest, especially when students appreciate the relevance of such activities to their future careers. On the other hand, attempts to provide extrinsic motivation using assessment, can actually push good students to adopt a surface approach to learning. This is discussed further under "Assessment" below.

7. Course design and delivery

Bowden and Marton (1998) questioned the aims of tertiary education. Based on their work, one might start a consideration of course design by asking if the aim of management education is.

a. To teach theories, formulas and methods of analysis,
b. To develop students' ways of thinking, or
c. To prepare leaders to meet unforeseeable challenges in an unknown future?

I would argue that we are trying to achieve all three of these objectives, both across an entire program of study and within each component (semester-long subject or course). The problem is to find the appropriate balance among these objectives. However, the first objective is the easiest to define, teach, and assess, and so the primary focus of course design is often on theories, formulas, and methods of analysis. The remaining objectives are harder to define, teach, and assess, but they are, arguably, at least as important as the first objective. In fact, objectives 2 and 3 incorporate the holy grail of "critical thinking", touted by some as the defining feature of a university-level course. Implicit in the third objective is developing professional competence including a commitment to lifelong learning. It becomes clear that course design and delivery involves a balance between the demands of meeting these objectives.

Once an initial decision has been made on the theories, formulas, and methods of analysis (objective 1) for a course, I have found that the remaining objectives can then be addressed by the choices of pedagogies, topic sequence, and assessment using the student learning literature as a guide (as discussed below). In particular, the use of experiential pedagogies such as the case method and industry-based projects, can develop ways of thinking from linear to multi-dimensional (see the section on sequencing, below). Such pedagogies also provide practice in dealing with unforeseen challenges. Note that the choice of pedagogy and sequencing may increase the time and/or the workload, so the decision on content may need to be revisited.

7.1. Pedagogies

The student learning literature has focused largely on the pedagogies that are common in traditional university courses – lectures, tutorials, set readings, essays, and sometimes laboratory classes. The pedagogies that are used in many schools of management, such as the case method, simulations, role plays and industry-based projects, have not been rigorously evaluated in terms of the student learning literature. However, when considered in the light of that literature, these student-centered pedagogies seem to meet many of the characteristics that are believed to foster a deep approach to learning (see, for example, Biggs et al., 2022, pp. 22–34; Ramsden, 2003, p. 80). There is also some empirical research that supports this view, at least for case method teaching (see, for example, Farashahi & Tajeddin, 2018; Rosier, 2022). There is scope for further research in this area, comparing pedagogies in terms of the student learning research, using the Revised Study Process Questionnaire (R-SPQ-2F) (Biggs et al., 2001) or the Approaches and Study Skills Inventory for Students (ASSIST) (Entwistle et al., 2013).

The incorporation of reflective exercises in management courses is already familiar to readers of this journal. The concept of reflection in education can be traced back at least to Dewey (1910). To borrow from the title of the book by Boud et al. (1985), reflection is the process of turning experience into learning. In particular, the work of Kolb (2014 – originally published in 1984) has influenced the use of reflection as part of experiential learning in management education. Several writers have proposed various exercises to promote reflection in management education (see, for example Daudelin, 1996; Kiersch; Gullekson, 2021; Rosier, 2002; Seibert, 1999; Waddock, 1999). Reflection has received significant attention in this journal with several hundred papers referring to the concept. What has received less attention is the link between reflection and approaches to learning. As noted above, a study by Leung and Kember (2003) found clear relationships between levels of reflection and approaches to learning. "Critical reflection" was found to be related to a deep approach to learning. It would seem that management educators who have incorporated some form of reflection into their courses, may have, in so doing, promoted a deep approach to learning in their students.

As this discussion indicates, some management educators are already using pedagogies that appear to foster a deep approach to learning. However, this is yet to be thoroughly tested empirically. With the exception of a few seemingly contradictory studies on the case method (see, for example, Ballantine et al., 2008; Rosier, 2022), none of these pedagogies have been tested in terms of approaches to learning.

However, the simple choice of pedagogies is just one step in the course design process, and just one of the factors that influence student approaches to learning. The benefits of effective pedagogy can be undermined or even lost by poor topic sequencing or inappropriate assessment.

7.2. Sequencing – Repetition and variation

Bowden and Marton (1998, 2014) and Marton and Trigwell (2000) argue that learning comes from discernment of a phenomenon, which in turn is only possible when that phenomenon varies in some way from the background. Marton and others have developed these ideas into what has become known as "variation theory" (Åkerlind, 2015; Lo & Marton, 2012; Marton, 2014; Pang & Marton, 2003, 2005).

If we think of a simple physical analogy, it is not easy to see a black cat on a dark night, but put that same cat in the middle of a snow-covered field on a sunny winter day, and it is easily seen. An observer can discern the shape, the colour and perhaps the movement of the cat because in all these respects, it varies from the background, which is flat, white, and not moving. However, this is only one aspect of variation and it provides limited information. If we want to truly understand the cat, we have to see it in a variety of situations – eating, sleeping, hunting, or playing. In these observations, the cat is constant and the situation (or what the cat is doing) is changing. Our attention (what Bowden and Marton (1998) call our "focal awareness") switches between the cat and the situation. It is the variation in backgrounds and discernment of the cat against these backgrounds, which enables us to better understand what a cat is.

In a learning situation, a student must simultaneously focus on the phenomenon being studied, and hold the background (prior knowledge) in peripheral awareness. Bowden and Marton use the reading of a text as an example. For instance, readers of this paper may be focusing on the ideas explained here, but at the same time they are aware of their previous experiences in management education and related fields. If a concept "strikes a chord" and they are able to relate an idea from this paper to a previously experienced concept or situation, then their focus changes to that previous situation. In other words, the previous situation becomes the focus considered against the background of this paper. Having considered the connection, readers will then refocus on the paper with their previous knowledge and experience in the background. It is this variation in background and foreground, combined with the refocusing on each, that enables understanding.

As an example, I introduce students to the concepts of throughput time, cycle time, and process capacity using the classic Harvard case "Kristen's Cookie Company" (Bohn, 1986) in which students are required to analyze the process for making cookies in a simple business. We also explore the ways in which throughput time, cycle time, and process capacity affect costs and hence profits. The concepts are seen as applicable to the production of goods in a simplified factory setting and students are easily able to apply the ideas to more complex production settings. In a later class, I use a different case to get students to apply the same concepts in the context of a restaurant – repeating the concepts, but varying the setting. Suddenly students start to realize that these concepts can be applied to a variety of service industries as well as to manufacturing. The concepts are then reinforced in a third case, in which they are applied to a fitness center. Similarly, most of the concepts we try to teach in management education can be initially seen in the context of a single case study or other exercise, but it is only when those same concepts are discerned against the changing backgrounds of several exercises, that students recognize their wider application.

Now let us consider variation in the context of a single, complex case study. Students have to consider all the interacting facts and concepts that make up the complex situation described. While most cases tend to highlight one academic discipline, it is usually necessary to consider multiple perspectives on the one case – what is the effect of a proposed change in product distribution on employees, how does it affect cash flow, what about the marketing implications, how will it affect the power structure in middle management, and so on. At or near the start of the class discussion, students will recognize the need to apply a particular concept to the case. At that point, the students' focus switches to the concept, with the case providing the context for the application of that concept. After that application, the students' awareness then shifts back to the case itself (which becomes the foreground with a range of concepts in the background) until another concept is seen as applicable. Then, again the application of that concept is the foreground against the background of the case. As students work on the case, they find that they have to be conscious of multiple concepts, with their focus shifting back and forth between the case and the concepts. Thus, we not only have variation in the application of each concept to a variety of cases, we also have variation in the concepts applied to each case. Students begin to appreciate the need to consider multiple perspectives on management situations.

There is a third type of variation that can be used in course design – the variation of pedagogy. The same management concept can be encountered in case studies, role plays, simulations, videos, and lectures. Variation in pedagogy, as practiced by some management educators, is another form of variation that may well assist learning. In fact, in one of my courses, the concepts of throughput time, cycle time, capacity, and the links to costs and profits (used in the example above) are also reinforced in a simulation conducted during class time.

Thus, variation theory guides us to plan an interplay of variation and repetition in our courses, all designed to assist students to appreciate the topic concepts in a variety of contexts and hence to appreciate both the generality and the limitations of the key concepts in a course. Empirical studies of the application of variation theory have been positive as reported by Åkerlind (2015, p. 17):

> "The impact of introducing Variation Theory into pedagogical design was typically dramatic, for example, 70% of the Variation Theory group versus 30% of the control group achieving the desired understanding of economic concepts (Pang & Marton, 2003, 2005) and 85% versus 25% achieving the desired understanding of accounting concepts (Rovio-Johansson, 2013; Rovio-Johansson & Lumsden, 2012). The impact on professional development of the teachers involved was also substantial (Pang, 2006; Åkerlind et al., 2011, 2014)."

Although these studies cited by Åkerlind were in business disciplines (economics and accounting), they were published in journals of education and higher education. Searches of the management education literature have failed to find any studies investigating the use of variation theory. When the use of variation theory resulted in the dramatic improvements in learning outcomes reported above, the absence of further studies of its use in business disciplines is remarkable.

7.3. Workload

High workload tends to drive students to adopt a surface approach to learning (Biggs et al., 2022, p. 24; Ramsden, 2003, p. 80). Thus, the workload must be manageable and this means there must be a limit to the content. Several writers have proposed a distinction between "threshold concepts" and "core concepts" to address this issue (Åkerlind et al., 2011; 2014; Biggs et al., 2022). Threshold concepts are concepts that "open up" new areas of understanding for students. Once understood, threshold concepts allow students to better appreciate a subject area and thus engage more deeply with the subject matter. Core concepts are important concepts in a subject, but do not open up entire new areas of understanding for students. In this paper, approaches to learning could be considered a threshold concept, while the effect of pedagogy on approaches to learning would be considered a core concept. Threshold concepts are the most important concepts to include in a course, whereas core concepts, although important, should be considered expendable if course content (and hence student workload) becomes excessive. If students really understand the threshold concepts, they are equipped to appreciate core concepts at a later date, even if those concepts were not explicitly included in the course.

After threshold concepts have been decided, other course content should be considered carefully, and culled if necessary, to ensure an appropriate workload for students. My own approach is to ask questions such as.

- Is it essential?
- Will it be covered in other subjects?
- Is it a fad or an enduring concept?

Perhaps more importantly, we should ask if inclusion of each item presents an opportunity to develop students' ways of thinking. Will it help prepare leaders to face unforeseen challenges in an unknown future? These purposes are usually best served by "going deeper" into a topic, especially a threshold concept, rather than packing more topics into a course.

7.4. Assessment

There is a great deal written about assessment in both the higher education literature and the management education literature, and there is no benefit in repeating it here except to highlight two serious issues – the anxiety often associated with assessment, and the effect of the format of assessment. The student learning literature is very clear about the effects of assessment on learning. The anxiety associated with assessment tends to drive students to adopt a surface approach to learning (Biggs, 1987; Biggs et al., 2022, p. 24; Boud, 1995; Fransson, 1977). The effects of assessment on learning were explored by Entwistle and Entwistle (2005) in a qualitative study, with high achieving students explaining how they were driven to resort to memorization in order to score well in examinations.

In fact, the format of assessments can send implicit messages to students about the approach to learning that they should adopt. Examinations that emphasize simple recall of course content are a clear signal that students should focus on memorizing content to do well in the exam. Good students who are interested in the subject, and are personally committed to achieving a good understanding of the course content, can be quite indignant about such simplistic forms of assessment. On the other hand, open book exams send the implicit message that understanding is essential. Such exams should be designed to test understanding, critique, and synthesis of concepts. Even better, they should require students to demonstrate the wise application of theories, concepts, and methods of analysis to realistic situations.

Given the problems associated with examinations, we should consider alternative forms of assessment. The obvious options come to mind – essays, written case study analyses, reflective journals, reflective reports, and term projects. If well-designed, these can be effective learning exercises as well as forms of assessment, but they all come with potential effects that can be detrimental to learning; anxiety and time pressure are the most obvious, especially when assignments are due in several courses at the same time.

Another possibility lies in the work of Dallimore et al. (2006) who found that the assessment of contributions to class discussion in a case method class, did not produce anxiety in students. While the reasons for this were not explored, it is possible that the unobtrusive nature of such in-class assessment, masked by lively discussion and perhaps even by the excitement of a good case method class, meant that students were not really aware of being assessed. It is also not clear whether this finding can be safely generalized outside the cultural setting of an American MBA program.

However, assessment is necessary in higher education for both formative and summative purposes. It is essential for accountability and as part of the process of evaluating course design and delivery. Thus, the design of assessment tasks requires management educators to balance several factors. This generally means minimizing the assessment load, but at the same time providing adequate opportunities for constructive feedback to students, and maintaining the rigor and integrity of assessment tasks.

8. Evaluation of course design and delivery

As noted above, the student learning research is already used in national surveys of course quality at universities in the UK (the National Student Survey) and Australia (the Student Experience Survey) with the results being made available publicly, to allow comparisons of course quality at participating institutions. A number of other surveys, based on the same research, are available to evaluate individual, semester-long subjects, or courses. The most frequently used of these survey instruments seem to be the Revised Study Process Questionnaire, (R-SPQ-2F) (Biggs et al., 2001) and the Approaches and Study Skills Inventory for Students (ASSIST) (Entwistle et al., 1997, 2013). The most obvious difference between these two surveys is that the R-SPQ-2F provides scores on two scales – surface approach and deep approach – whereas the ASSIST provides scores on the same two scales plus a score on a strategic approach. Remember that a high score on the strategic scale indicates a willingness to change between surface and deep approaches according to the circumstances. Before using either of these surveys, it is important to have a clear idea of what we want to measure and how we understand the concept of approaches to learning. Are approaches to learning innate characteristics of students, which may change slowly over time in response to educational experiences, contexts, changing interests and other factors, or are they a response to a particular educational situation?

If we are trying to measure the effect of a change in course design or pedagogy, then we want to measure students' responses to that change, in terms of surface and deep approaches, and it is measures of these scores that we are interested in. Measuring scores on a strategic approach would add little or nothing to our understanding of the effect of the intervention. For this reason, the shorter R-SPQ-2F is preferred for such evaluations. However, the survey should specifically relate to the intervention, changing wording to suit the application, as recommended by the developers, so that it is clear to participants that they are to provide answers based on their response to the particular intervention or pedagogy being investigated.

There have been some studies, where these or similar survey instruments were used to evaluate the effects of interventions in business courses, especially in accounting ((see, for example, Ballantine et al., 2008; English et al., 2004; Hall et al., 2004). However, the results of some of these studies were contradictory, and a careful reading of these papers raises questions about uncontrolled variables, such as concurrent courses that were not modified, and the design of the evaluation to ensure that the effect of the intervention was what was actually being measured.

Management education makes use of pedagogies such as the case method and simulations that are less commonly used in other disciplines, and these pedagogies can and should be evaluated in terms of the student learning research. Such studies need very careful design and control if the evaluation is to be considered valid.

9. Implications for practice

The student learning research provides a research-based approach to improving higher education, and its findings are arguably just as valid for management education. While many management educators use apparently effective pedagogies, the student learning research provides an empirical basis for course design, and for selection of pedagogies, topic sequencing, and methods of assessment. However, business schools should be cautious in the way this research is applied. I have seen cases where policies have been imposed based on a superficial understanding of the research. At best, these approaches have had no real effect on teaching practice; at worst, they have acted as constraints on course improvement. Rather than imposing such a top-down approach, it is better that faculty actually read and understand the student learning research, and work together to apply its findings in the unique context of management education.

Kandlbinder's study (2013) provides a quick overview of the student learning research. A summary of the original research, written by the original researchers, can be found in Marton et al. (2005). At the time of writing, this was available online at the University of Edinburgh website. Chapter three by Marton and Säljö provides a good introduction to the research. The textbook on university teaching by Biggs et al. (2022) provides a comprehensive overview of the implications of the student learning research. Trigwell and Prosser (2020) provide a more succinct summary of the research and its implications.

Given the remarkable results achieved through the application of variation theory (noted above), management educators could give special attention to the work of Åkerlind (2015), Pang and Marton (2003, 2005), Rovio-Johansson (2013), and Rovio-Johansson and Lumsden (2012). Careful planning of the interplay of repetition and variation in the sequencing of learning activities, has the potential to "dramatically" improve learning outcomes.

However, management educators will find little reference to pedagogies such as case method teaching or the use of simulations, in the student learning literature. While we can make assumptions about applying the student learning research to such pedagogies, those assumptions should be tested. The R-SPQ-2F, noted above, can be used for this purpose (Biggs et al., 2001).

As management education evolves, with increased use of online courses, blended courses (combining online and face-to-face), online simulations, video cases, and other innovations, an understanding of the student learning research will enable management educators to make better choices in course design and delivery. Again, the R-SPQ-2F can be used to evaluate the choices made.

10. Future research

The application of findings, concepts, and methods of the student learning research to management education presents opportunities for further research in management education. Further empirical research on the case method and the use of simulations are perhaps the most obvious gaps in the student learning literature. Understanding the effects of various forms of assessment on student learning in management education would also be valuable. The results of applying variation theory, noted by Åkerlind (2015) call for further investigation of this approach to course design. Other opportunities for further research have been noted throughout this paper.

Finally, as noted above, the student learning literature provides simple surveys such as the Revised Study Process Questionnaire (R-SPQ-2F) (Biggs et al., 2001) and the Approaches to Teaching Inventory (ATI) (Trigwell & Prosser, 2004; Trigwell et al., 2005). These can be used in simple but valuable research projects to evaluate pedagogies, learning activities, course design and even staff development programs.

11. Conclusion

In this paper, I have presented an introduction to the student learning research and its potential application to management education. The student learning research has had a significant impact in higher education generally, especially in Britain, Europe, Hong Kong, and Australia. However, it has received less attention in the field of management education. It offers research-based guidelines for course design and evaluation as well as simple, proven surveys that can be used in management education research. Because many of the pedagogies used in management education, especially the case method, simulations, and industry-based projects have not been thoroughly investigated in terms of the student learning research, there is scope for significant further research in these areas.

In summary, addressing the gaps between the higher education research and management education, opens new possibilities for the research-based development of management education practice and new areas for management education research.

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